Algebra and Functions

Chapter 1 · Original slide 4 · 1: Index Laws

base

3^5

exponent or index (plural: indices)

!

$a^m \times a^n = a^{m+n}$
$a^m \div a^n = a^{m-n}$
$(a^m)^n = a^{mn}$
$(ab)^n = a^n b^n$

power

Some also refer to the 5 as the ‘power’. But this would lead to unfortunate ambiguous phrases like “when we multiply two powers together, we add the powers” instead of “when we multiply two powers together, we add the indices”.

Simplify $\left(a^3\right)^2 \times 2a^2$

Simplify $\left(4x^3y\right)^3$

Simplify $2x^2\left(3+5x\right) - x\left(4-x^2\right)$

Simplify $\frac{x^3-2x}{3x^2}$

Tip: A common student error is to get the sign wrong of $+x^3$

Tip: While $\frac{a+b}{c}$ can be split into $\frac{a}{c} + \frac{b}{c}$, a common student error is to think that $\frac{a}{b+c} = \frac{a}{b} + \frac{a}{c}$

Chapter 1 · Original slide 5 · Test Your Understanding

1

Simplify $\left(\frac{2a^5}{a^2}\right)^2 \times 3a$

2

Simplify $\frac{2x+x^5}{4x^3}$

3

Expand and simplify
$2x\left(3-x^2\right) - 4x^3\left(3-x\right)$

Note: This is using $(ab)^n = a^n b^n$ law backwards.

4

Simplify $2^x \times 3^x$

Chapter 1 · Original slide 6 · Exercise 1A – Pearson Pure Y1 page 4

2 Expand and simplify if possible:

o $x(3x^2 - 2x + 5)$
p $7y^2(2 - 5y + 3y^2)$
q $-2y^2(5 - 7y + 3y^2)$
r $7(x - 2) + 3(x + 4) - 6(x - 2)$
s $5x - 3(4 - 2x) + 6$
t $3x^2 - x(3 - 4x) + 7$
u $4x(x + 3) - 2x(3x - 7)$
v $3x^2(2x + 1) - 5x^2(3x - 4)$

d $\frac{8x^3+5x}{2x}$ **e** $\frac{7x^7+5x^2}{5x}$ **f** $\frac{9x^5-5x^3}{3x}$

3 Simplify these fractions:

Chapter 1 · Original slide 7 · Exercise 1A – Pearson Pure Y1 page 4

Extension (Full Database: http://www.drfrostmaths.com/resources/resource.php?rid=268)

2

[MAT 2012 1B] Let $N = 2^k \times 4^m \times 8^n$ where k,m,n are positive whole numbers. Then N will definitely be a square number whenever:

k is even;

$k + n$ is odd;

k is odd but $m + n$ is even;

$k + n$ is even.

1

[MAT 2006 1A] Which of the following numbers is largest?

$$\left((2^3)^2\right)^3, \quad (2^3)^{(2^3)}, \quad 2^{((3^2)^3)}, \quad 2^{(3^{(2^3)})}$$

Chapter 1 · Original slide 17 · 4: Negative and Fractional Indices

$a^0 = 1$
$a^{\frac{1}{m}} = \sqrt[m]{a}$
$a^{\frac{n}{m}} = \left(\sqrt[m]{a}\right)^n$
$a^{-m} = \frac{1}{a^m}$

Prove that $x^{\frac{1}{2}} = \sqrt{x}$

Evaluate $32^{\frac{2}{5}}$

Evaluate $\left(\frac{27}{8}\right)^{-\frac{2}{3}}$

Note: $\sqrt{9}$ only means the positive square root of 9, i.e. 3 not -3.

Otherwise, what would be the point of the $\pm$ in the quadratic formula before the $\sqrt{b^2 - 4ac}$?

Evaluate $27^{-\frac{1}{3}}$

Simplify $\left(\frac{1}{9}x^6y\right)^{\frac{1}{2}}$

If $b = \frac{1}{9}a^2$, determine $3b^{-2}$ in the form kb^n where k, n are constants.

Chapter 1 · Original slide 18 · Writing a surd using indices

If $9\sqrt{3} = 3^k$, determine the value of k.

Note: In algebra we like to avoid mixed numbers. So don't write $3^{2\frac{1}{2}}$

The key here is to write everything as powers with a consistent base, in this case, 3.

Chapter 1 · Original slide 19 · Further Examples

[Edexcel IGCSE May14-4H] Given that $\left(2^{\frac{1}{2}}\right)^n = \frac{2^x}{8^y}$

Express n in terms of x and y.

Write 8 as a power of 2 (putting a bracket around it), for consistency of base with the other powers.

[Edexcel IAL C12 Jan 2019 Q2c] Given that $y = 2^x$, express the following in terms of y.

$\frac{1}{4^{2x-3}}$

Write your expression in its simplest form.

Chapter 1 · Original slide 20 · Test Your Understanding

Given $2^x \times 4^y = \frac{1}{2\sqrt{2}}$ express y as a function of x. (3)

Chapter 1 · Original slide 21 · Exercise 1D

4 a Find the value of $81^{\frac{1}{4}}$. **b** Simplify $x(2x^{-\frac{1}{3}})^4$.

5 Given that $y = \frac{1}{8}x^3$ express each of the following in the form kx^n, where k and n are constants.
a $y^{\frac{1}{3}}$ **b** $\frac{1}{2}y^{-2}$

Challenge

Chapter 1 · Original slide 22 · 5: Surds

Recap:

Note: A *rational* number is any which can be expressed as $\frac{a}{b}$ where a, b are integers. $\frac{2}{3}$ and $\frac{4}{1} = 4$ are rational numbers, but π and $\sqrt{2}$ are not.

A surd is a root of a number that does not simplify to a rational number.

Laws:

$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$
$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$

$\sqrt{3} \times 2$
$3\sqrt{5} \times 2\sqrt{5}$
$\sqrt{8} = \sqrt{4}\sqrt{2}$
$\sqrt{12} + \sqrt{27}$
$\left(\sqrt{8} + 1\right)\left(\sqrt{2} - 3\right)$

Chapter 1 · Original slide 23 · Exercise 1E – Pure Y1 Page 13

3 Simplify $\sqrt{75} - \sqrt{12}$ giving your answer in the form $a\sqrt{3}$, where a is an integer.

Extension

1

[SMC 2014 Q24] Which of the following is smallest?

$10 - 3\sqrt{11}$; $8 - 3\sqrt{7}$; $5 - 2\sqrt{6}$; $9 - 4\sqrt{5}$; $7 - 4\sqrt{3}$.

2

[SMC 2012 Q21] Which of the following numbers does not have a square root in the form $x + y\sqrt{2}$, where x and y are positive integers?

$17 + 12\sqrt{2}$; $22 + 12\sqrt{2}$; $38 + 12\sqrt{2}$; $54 + 12\sqrt{2}$; $73 + 12\sqrt{2}$.

Chapter 1 · Original slide 24 · 6: Rationalising The Denominator

Here's a surd. What could we multiply it by such that it's no longer an irrational number?

$\sqrt{5}\times$

$\frac{1}{\sqrt{2}}$

In this fraction, the denominator is irrational. 'Rationalising the denominator' means making the denominator a rational number.

What could we multiply this fraction by to both rationalise the denominator, but leave the value of the fraction unchanged?

Side Note: There's two reasons why we might want to do this:

1. For aesthetic reasons, it makes more sense to say "half of root 2" rather than "one root two-th of 1". It's nice to divide by something whole!

2. It makes it easier for us to add expressions involving surds.

Chapter 1 · Original slide 25 · Examples

$\frac{3}{\sqrt{2}} =$

$\frac{6}{\sqrt{3}} =$

$\frac{7}{\sqrt{7}} =$

$\frac{15}{\sqrt{5}} + \sqrt{5} =$

Chapter 1 · Original slide 26 · More Complex Denominators

You've seen 'rationalising a denominator', the idea being that we don't like to divide things by an irrational number.

We basically use the same expression but with the sign reversed (this is known as the *conjugate*). That way, we obtain the difference of two squares. Since $(a+b)(a-b) = a^2 - b^2$, any surds will be squared and thus we'll end up with no surds in the denominator.

But what do we multiply the top and bottom by if we have a more complicated denominator?

$\frac{1}{\sqrt{2}+1} \times$

Chapter 1 · Original slide 27 · More Examples

$\frac{3}{\sqrt{6}-2} \times$

You can explicitly expand out $\left(\sqrt{6}-2\right)\left(\sqrt{6}+2\right)$ in the denominator, but remember that $(a-b)(a+b)=a^2-b^2$ so we can mentally obtain $6-4=2$

Just remember: 'difference of two squares'!

$\frac{4}{\sqrt{3}+1} \times$

$\frac{3\sqrt{2}+4}{5\sqrt{2}-7} \times$

Chapter 1 · Original slide 28 · Test Your Understanding

Rationalise the denominator and simplify

$\frac{4}{\sqrt{5}-2}$

Solve $y\left(\sqrt{3}-1\right)=8$

Give your answer in the form $a+b\sqrt{3}$ where a and b are integers.

Rationalise the denominator and simplify

$\frac{2\sqrt{3}-1}{3\sqrt{3}+1}$

Chapter 1 · Original slide 29 · Writing surd expressions in power form

This is not in the textbook, but a common type of question in exams, often asked in conjunction with differentiation (a later chapter).

Express $\frac{(2+\sqrt{x})^2}{x}$ as powers of x.

Write any roots as powers of x.

Test Your Understanding

Express $\frac{(x+\sqrt{x})^2}{2\sqrt{x}}$ as powers of x.

Expand.

Split fraction (some may wish to do this step mentally)

Chapter 1 · Original slide 30 · Exercise 1F (Page 15)

or alternatively: (not in textbook)

2 3 4 5 6

Expand and simplify:

$\left(\sqrt{5}+3\right)\left(\sqrt{5}-2\right)\left(\sqrt{5}+1\right) =$

Rationalise the denominator, giving your answer in the form a+b√3.

$\frac{3\sqrt{3}+7}{3\sqrt{3}-5} =$

Solve x(4−√6)=10 giving your answer in the form a+b√6.

Solve y(1+√2)−√2=3

Simplify:

$\frac{\sqrt{a+1}-\sqrt{a}}{\sqrt{a+1}+\sqrt{a}} =$

1 a b c d e

Rationalise the denominator and simplify the following:

$\frac{1}{\sqrt{5}+2} =$

$\frac{\sqrt{3}}{\sqrt{3}-1} =$

$\frac{\sqrt{5}+1}{\sqrt{5}-2} =$

$\frac{2\sqrt{3}-1}{3\sqrt{3}+4} =$

$\frac{5\sqrt{5}-2}{2\sqrt{5}-3} =$

Chapter 2 · Original slide 5 · 1: Solving Quadratic Equations

$x^2 + 5x = 6$

There are three ways of solving a quadratic equation.

One is by *completing the square*, which we'll do later.

What are the other two ways?

By factorisation

Using the quadratic formula.

If $ax^2 + bx + c = 0$ then

$x^2 + 5x - 6 = 0$

Put in form.

$ax^2 + bx + c = 0$ with 0 on one side

Factorise. If the product of two things is 0, at least one must be 0.

Chapter 2 · Original slide 6 · 1: Solving Quadratic Equations

Solving without factorising

If the subject only appears once however, it might be easier not to expand out/factorise:

If you can't see why the $\pm$ is required, think about the solutions to: $x^2 = 4$.

$2^2 = 4$, but $(-2)^2 = 4$ as well!

So $x = \pm 2$.

Quadratics 'in disguise'

When we have an expression like say $x^2 + 3x - 2$, we say it is "quadratic in x". In trigonometry you will have to solve equations like $(\sin x)^2 + 3\sin x + 2$. We say that the expression is "quadratic in $\sin x$".

Either use a suitable substitution so that you have a 'normal' quadratic, or go straight for the factorisation if you're feeling more confident (recommended!).

Solve $x - 6\sqrt{x} + 8 = 0$

substitution

hardcore

Chapter 2 · Original slide 7 · Test Your Understanding

1
Solve $(x+3)^2 = x+5$ using factorisation.

2
Solve $(2x+1)^2 = 5$

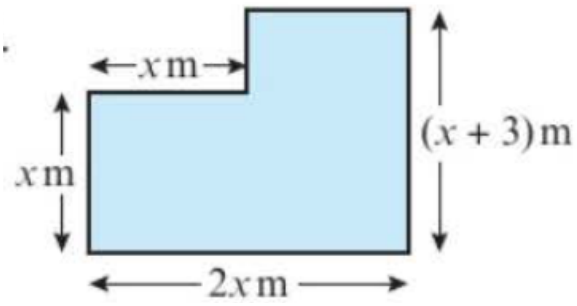
3
Solve $\sqrt{x+3} = x-3$

4
Solve $2x + \sqrt{x} - 1 = 0$

"Use of Technology":

The 'equation' mode on your calculator will solve quadratics in the form $ax^2 + bx + c = 0$. When you're asked for the 'order', use 2 (we'll see why later).

Chapter 2 · Original slide 8 · Exercise 2A/2B Pure Y1 Pages 20-22



3 Solve the following equations:

i $6x^2 - 7 = 11x$

h $5x^2 - 10x^2 = -7 + x + x^2$

4 This shape has an area of $44\,\text{m}^2$. Find the value of x.

5 Solve the equation $5x + 3 = \sqrt{3x+7}$.

Chapter 2 · Original slide 9 · Exercise 2A/2B Pure Y1 Pages 20-22

2 Solve the following equations using the quadratic formula. Give your answers to three significant figures.

h $2x^2 + 11x = 5x^2 - 18$

3 For each of the equations below, choose a suitable method and find all of the solutions. Where necessary, give your answers to three significant figures.

e $(2x + 8)^2 = 100$ **f** $6x^2 + 6 = 12x$

Chapter 2 · Original slide 10 · Exercise 2A/2B Pure Y1 Pages 20-22

1

Extension: (Full Database: http://www.drfrostmaths.com/resources/resource.php?rid=268)

Chapter 2 · Original slide 11 · 2: Completing the Square

“Completing the square” means putting a quadratic in the form $(x+a)^2+b$

or $a(x+b)^2+c$

The underlying reason we do this is because $\mathbf{x}$ only appears once in the expression (e.g. in $(x+2)^2+3$ vs x^2+4x+7), which makes it algebraically easier to handle. This has a number of consequences:

a. Solving Quadratics

b. Sketching Quadratics

If we have a completed square:

$(x+4)^2-7=0$

we saw at the start of the chapter how we could rearrange to make x the subject.

Indeed using the quadratic formula is actually solving the quadratic by completing the square – it’s just someone has done the work for us already!

We’ll see later that if $y=(x+a)^2+b$, then the minimum point is $(-a,b)$

c. In integration

In Further Maths, completing the square allows us to ‘integrate’ expressions like:

$\int \frac{1}{x^2-4x+5}dx$

(you will cover integration later this module)

Chapter 2 · Original slide 12 · Completing the Square Recap

Textbook Note:

The textbook uses the formula

$x^2+bx=\left(x+\frac{b}{2}\right)^2-\left(\frac{b}{2}\right)^2$

and similarly

$ax^2+bx+c=a\left(x+\frac{b}{2a}\right)^2+\left(c-\frac{b^2}{4a}\right)$

My personal judgement is that it’s not worth memorising these and you should instead think through the steps. Even the textbook agrees!

Expand:

$(x+\mathbf{9})^2=$
$(x-\mathbf{5})^2=$

What do you notice about the relationship between the bold numbers?

Therefore if we had x^2+12x, how could we write it in the form $(x+a)^2+b$?

Since $(x+6)^2=x^2+12x+36$, we want to discard the 36, so ‘throw it away’ by subtracting.

Further Examples:

$x^2+8x=$
$x^2-2x=$
$x^2-6x+7=$

Notice that despite the a being negative, we still subtract after the bracket as $(-1)^2$ is positive.

Chapter 2 · Original slide 13 · Completing the Square

Express $2x^2 + 12x + 7$ in the form $a(x + b)^2 + c$

Factorise out coefficient of x^2. You can leave the constant term outside the bracket.

Complete the square inside the bracket (you should have two sets of brackets)

Expand out outer bracket.

Express $5 - 3x^2 + 6x$ in the form $a - b(x + c)^2$

It may help to write in the form $ax^2 + bx + c$ first.

Chapter 2 · Original slide 14 · Test Your Understanding

Express $3x^2 - 18x + 4$ in the form $a(x + b)^2 + c$

Express $20x - 5x^2 + 3$ in the form $a - b(x + c)^2$

Chapter 2 · Original slide 15 · Solving by Completing the Square

Solve the equation:

$3x^2 - 18x + 4 = 0$

Reflections: Previously we factorised out the 3. This is because $3x^2 - 18x + 4$ on its own is an expression, so dividing by 3 (instead of factorising) would change the expression.

However, in an equation, we can divide both sides by 3 without affecting the solutions.

Chapter 2 · Original slide 16 · Exercise 2C/2D Pure Y1 Page 23-24

1 Complete the square for the expressions:

a $x^2 + 4x$
b $x^2 - 6x$
c $x^2 - 16x$
d $x^2 + x$
e $x^2 - 14$

2 Complete the square for the expressions:

a $2x^2 + 16x$
b $3x^2 - 24x$
c $5x^2 + 20x$
d $2x^2 - 5x$
e $8x - 2x^2$

Chapter 2 · Original slide 17 · Exercise 2C/2D Pure Y1 Page 23-24

4 Given that $x^2 + 3x + 6 = (x + a)^2 + b$ find the values of the constants a and b

5 Write $2 + 0.8\text{x} - 0.04x^2$ in the form $A - B(x + C)^2$ where $A \text{and} B$are constants to be determined

4 By completing the square, show that the solutions to the equation $x^2 + 2bx + c = 0$ are given by the formula $x = -b \pm \sqrt{b^2 - c}$. (4 marks)

Chapter 2 · Original slide 18 · Proving the Quadratic Formula

If $ax^2 + bx + c = 0$, prove that $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Just use exactly the same method as you usually would!

Chapter 2 · Original slide 24 · Test Your Understanding

1

Find the minimum value of $f(x) = 2x^2 + 12x - 5$ and state the value of x for which this occurs.

2

Find the roots of the function $f(x) = 2x^2 + 3x + 1$

3

Find the roots of the function $f(x) = x^4 - x^2 - 6$

Chapter 2 · Original slide 26 · 4: Quadratic Graphs

Recall that x refers to the input of a function, and the expression $f(x)$ refers to the output. For graph sketches, we often write $y = f(x)$, i.e. we set the y values to be the output of the function.

y

A 'sketch' in maths has a clearly defined meaning:

• We care about the general shape of the graph, not exact points.

• The axes should have no scale.

• Generally the only coordinates indicated are intercepts with the axis or other points of interest (e.g. intersections of multiple graphs)

Features needed in sketch of quadratic?

Recall a root of a function is where the output, in this case the y value, is 0.

Roots

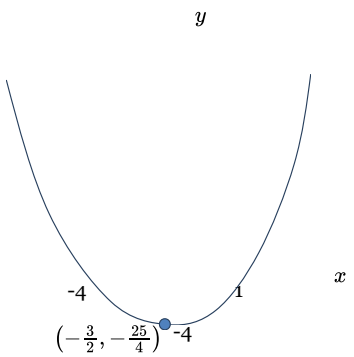
x

General shape: Smiley face or hill?

y-intercept

Turning point

Chapter 2 · Original slide 27 · Example



Sketch the graph of $y = x^2 + 3x - 4$ and find the coordinates of the turning point.

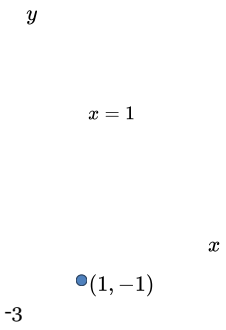
Roots:

Turning point:

y-intercept:

Recall that if $f(x) = (x + a)^2 + b$, the minimum output is b and $-a$ is the x value which minimises it. i.e. Turning point is $(-a, b)$

Chapter 2 · Original slide 28 · Example



Sketch the graph of $y = 4x - 2x^2 - 3$ and find the coordinates of the turning point. Write down the equation of the line of symmetry.

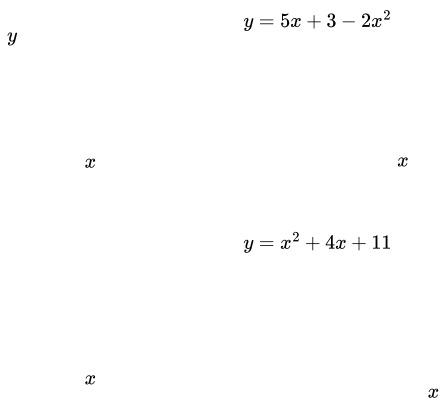
Roots:

Turning point:

y-intercept:

Line of reflection:

Chapter 2 · Original slide 29 · Test Your Understanding



Sketch the following, indicating any intercepts with the axis, the turning point and the equation of the line of symmetry.

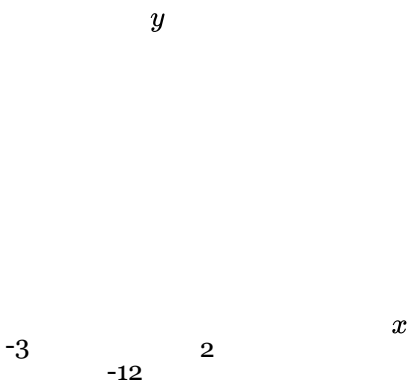
a c

$y = x^2 + 4$

b d

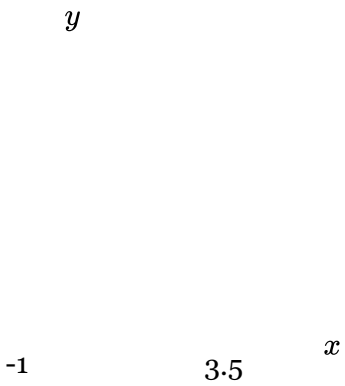
$y = x^2 - 7x + 10$

Chapter 2 · Original slide 30 · Determining the Equation using a Graph



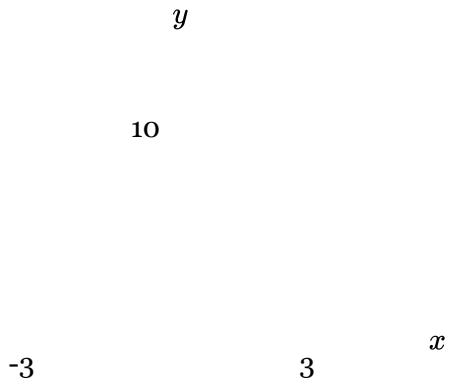
Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$.

Chapter 2 · Original slide 31 · Further Example



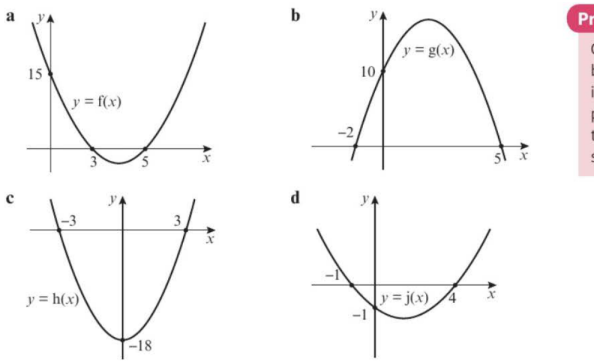
Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$, where a, b, c are integers.

Chapter 2 · Original slide 32 · Test Your Understanding



Determine an equation of this quadratic graph.

Chapter 2 · Original slide 33 · Exercise 2F Pure Y1 Page 30



2 These sketches are graphs of quadratic functions of the form $ax^2 + bx + c$. Find the values of a,b,c for each function.

Chapter 2 · Original slide 34 · Exercise 2F Pure Y1 Page 30

3 The graph of $y = ax^2 + bx + c$ has a minimum at (5,-3) and passes through (4,0). Find the values of a,b,c.

[MAT 2003 1H] Into how many regions is the plane divided when the following three parabolas are drawn?

$y = x^2$
$y = x^2 - 2x$
$y = x^2 + 2x + 2$

x

Extension Question:

y

Chapter 2 · Original slide 35 · Starter

How many distinct real solutions do each of the following have?

$x^2 - 12x + 36 = 0$

$x^2 + x + 3 = 0$

$x^2 - 2x - 1 = 0$

Chapter 2 · Original slide 36 · 5: The Discriminant

$ax^2 + bx + c = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Note: Roots of a function f are the values of x such that $f(x) = 0$.

Similarly the roots of an equation are solutions to an equation in the form $f(x) = 0$

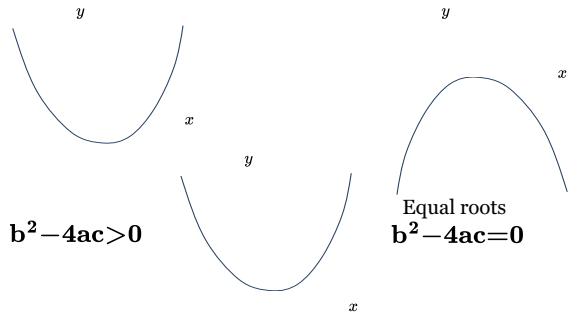
Looking at this formula, when in general do you think we have:

- No real roots?
- Equal roots?
- Two distinct roots?

$b2 - 4ac$is known as the discriminant.

Because adding 0 or subtracting 0 in the quadratic formula gives the same value.

Chapter 2 · Original slide 37 · 5: The Discriminant



No real roots

We say the graph “touches” the axis.

Distinct real roots

$b^2-4ac<0$

Chapter 2 · Original slide 38 · Just for your interest…

Why do we say “Equal Roots” not “One root”?

It is due to the Fundamental Theorem of Algebra:

“Every polynomial of order n has exactly n roots.”

A polynomial is an expression with non-negative integer powers of x, i.e.

$a + bx + cx^2 + dx^3 + \ldots$

All linear, quadratic and cubic expressions are examples of polynomials.

$x^2 - 12x + 36 = 0$

Using the quadratic formula gives us the same value in both $+$ and $-$ cases: $x = 6$.

You might wonder why we say “*it has one repeated root*” or “*it has equal roots*”, i.e. indicating we have 2 roots (but with the same value). Why not say it has 1 root?

Despite the theorem being a simple statement, it was only until 1806 that it was first proven by Argand. Clearly by using the quadratic formula we can show a quadratic equation has 2 roots. We can use similar formulas to show that a cubic has 3 roots and a quartic 4 roots. But there is provably no such formulae for order 5 (quintics) and beyond. So we have to prove for example that 5 roots exist for a quintic, despite us having no way to find these exact roots!

One side result of the Fundamental Theorem of Algebra is that every polynomial can be written as a product of linear and/or quadratic expressions.

Leibniz claimed in 1702 that a polynomial of the form $x^4 + a^4$ cannot be written in this way. He then got completely burned by Euler in 1742 who managed to do so:

$$x^4 + a^4 = \left(x^2 + a\sqrt{2}x + a^2\right)\left(x^2 - a\sqrt{2}x + a^2\right)$$

The order of a polynomial is its highest power of x. So the order of a quadratic is 2, and a cubic 3.

Chapter 2 · Original slide 39 · Quickfire Questions

Equation	Number of Distinct Real Roots
$x^2 - 4x + 1 = 0$	
$x^2 - 4x + 4 = 0$	
$2x^2 - 6x - 3 = 0$	
$x - 4 - 3x^2 = 0$	
$1 - x^2 = 0$	

Equation	Discriminant
$x^2 + 3x + 4 = 0$	

Chapter 2 · Original slide 40 · Problems involving the discriminant

8. The equation $x^2 + 2px + (3p + 4) = 0$, where p is a positive constant, has equal roots.

a)

(a) Find the value of p. (4)

(b) For this value of p, solve the equation $x^2 + 2px + (3p + 4) = 0$. (2)

Tip: Always start by writing out a, b and c explicitly.

b)

Chapter 2 · Original slide 41 · Exercise 2G Pure Y1 Page 32

2 Find the values of k for which $x^2 + 6x + k = 0$ has two real solutions.

3 Find the value of t for which $2x^2 - 3x + t = 0$ has exactly one solution.

Chapter 2 · Original slide 42 · Exercise 2G Pure Y1 Page 32

4 Given that the function $\mathrm{f}(x) = sx^2 + 8x + s$ has equal roots, find the value of the positive constant s.

5 Find the range of values of k for which $3x^2 - 4x + k = 0$ has no real solutions.

Chapter 2 · Original slide 43 · Exercise 2G Pure Y1 Page 32

7 $h(x) = 2x^2 + (k + 4)x + k$, where k is a real constant.

a Find the discriminant of $h(x)$ in terms of k.

b Hence or otherwise, prove that $h(x)$ has two distinct real roots for all values of k.

6 The function $g(x) = x^2 + 3px + (14p - 3)$, where p is an integer, has two equal roots.

a Find the value of p.

b For this value of p, solve the equation $x^2 + 3px + (14p - 3) = 0$.

Chapter 2 · Original slide 44 · Test Your Understanding

$x^2 + 5kx + (10k + 5) = 0$

where k is a constant.

Given that this equation has equal roots, determine the value of k.

Find the range of values of k for which $x^2 + 6x + k = 0$ has two distinct real solutions.

Chapter 2 · Original slide 45 · Exercise 2G Pure Y1 Page 32

Extension Questions:

1

[MAT 2009 1C] Given a real constant c, the equation

$x^4 = (x - c)^2$

Has four real solutions (including possible repeated roots) for:

A) $c \leq \frac{1}{4}$

B) $-\frac{1}{4} \leq c \leq \frac{1}{4}$

C) $c \leq -\frac{1}{4}$

D) all values of c

2

[MAT 2006 1B] The equation $\left(2 + x - x^2\right)^2 = 16$ has how many real root(s)?

3

[MAT 2011 1B] A rectangle has perimeter P and area A. The values P and A must satisfy:

A) $P^3 > A$

B) $A^2 > 2P + 1$

C) $P^2 \geq 16A$

D) $PA > A + P$

Chapter 2 · Original slide 46 · Modelling

The new A Level has a particular emphasis of the application of theory to real-life situations. A mathematical model is the maths used to model such a situation, possibly with some simplifying assumptions.

Example (from textbook): A spear is thrown over level ground from the top of a tower.

The height, in metres, of the spear above the ground after t seconds is modelled by the function:
$h(t) = 12.25 + 14.7t - 4.9t^2, t \geq 0$

a) Interpret the meaning of the constant term 12.25 in the model.

b) After how many seconds does the spear hit the ground?

c) Write $h(t)$ in the form $A - B(t - C)^2$, where A, B and C are constants to be found.

d) Using your answer to part c or otherwise, find the maximum height of the spear above the ground, and the time at which this maximum height is reached?

Chapter 2 · Original slide 47 · Exercise 2H Pure Y1 Page 34

2 A car manufacturer uses a model to predict the fuel consumption, y miles per gallon (mpg), for a specific model of car travelling at a speed of x mph.

$$y = -0.01x^2 + 0.975x + 16,\ x > 0.$$

a Use the model to find two speeds at which the car has a fuel consumption of 32.5 mpg.
b Rewrite y in the form $A - B(x - C)^2$, where A, B, C are constants to be found.
c Using your answer to part b, find the speed at which the car has the greatest fuel efficiency.
d Use the model to calculate the fuel consumption of a car travelling at 120 mph.
Comment on the validity of using this model for very high speeds.

Chapter 2 · Original slide 48 · Exercise 2H Pure Y1 Page 34

3 A fertiliser company uses a model to determine how the amount of fertiliser used, f kilograms per hectare, affects the grain yield g, measured in tonnes per hectare.

$$g = 6 + 0.03f - 0.00006f^2.$$

a According to the model, how much grain would each hectare yield without any fertiliser? (1 mark)
b One farmer currently uses 20 kilograms of fertiliser per hectare. How much more fertiliser would he need to use to increase his grain yield by 1 tonne per hectare? (4 marks)

Chapter 2 · Original slide 49 · Exercise 2H Pure Y1 Page 34

4 A football stadium has 25000 seats. The football club know from past experience that they will sell only 10000 tickets if each ticket costs \pounds30. They also expect to sell 1000 more tickets every time the price goes down by \pounds1.

a The number of tickets sold t can be modelled by the linear equation $t = M - 1000p$, where \pounds$p$ is the price of each ticket and M is a constant. Find the value of M. (1 mark)

The total revenue, \pounds$r$, can be calculated by multiplying the number of tickets sold by the price of each ticket. This can be written as $r = p(M - 1000p)$.

b Rearrange r into the form $A - B(p - C)^2$, where A, B, C are constants to be found. (3 marks)

c Using your answer to part b or otherwise, work out how much the football club should charge for each ticket if they want to make the maximum amount of money. (2 marks)

Chapter 3 · Original slide 3 · Solutions sets

The solution(s) to an equation may be:

$2x + 1 = 5$

$x^2 + 3x + 2 = 0$

$x > 3$

$x^2 = -1$

$x^2 + x = x\,(x + 1)$

! The solutions to an equation are known as the solution set.

A single value:

Multiple values:

An infinitely large set of values:

No (real) values!

Every value!

The point is that you shouldn't think of the solution to an equation/inequality as an 'answer', but a set of values, which might just be a set of 1 value (known as a singleton set), a set of no values (i.e. the empty set $\varnothing$), or an infinite set (in the last example above, this was $\mathbb{R}$)

Chapter 3 · Original slide 4 · Solutions sets

Scenario	Example	Solution Set
	$x + y = 9$ $x - y = 1$	
No solutions:	$x + y = 1$ $x + y = 3$	
	$x + y = 1$ $2x + 2y = 2$	

For simultaneous equations, the same is true, except each ‘solution’ in the solution set is an assignment to multiple variables.

All equations have to be satisfied at the same time, i.e. ‘simultaneously’.

A single solution:

$x^2 + y^2 = 10$
$x + y = 4$

Two solutions:

Infinitely large set of solutions:

Chapter 3 · Original slide 5 · 1 :: Simultaneous Equations

Recap!

Solve the simultaneous equations:

$3x + y = 8$
$2x - 3y = 9$

Solve the simultaneous equations:

$x + 2y = 3$
$x^2 + 3xy = 10$

We can’t use elimination this time as nothing would cancel.

We instead:

(1) Rearrange linear equation to make x or y the subject.

(2) Substitute into quadratic equation and solve.

We can either use substitution (i.e. making x or y the subject of one equation, and substituting it into the other) or elimination, but the latter is easier for linear equations.

Chapter 3 · Original slide 6 · Test Your Understanding

Solve the simultaneous equations:

$3x^2 + y^2 = 21$
$y = x + 1$

Chapter 3 · Original slide 7 · Exercise 3A/B – Y1 Pages 40, 41

4 Solve the simultaneous equations:

$x + y = 3, \quad x^2 - 3y = 1$

Chapter 3 · Original slide 8 · Exercise 3A/B – Y1 Pages 40, 41

5 a By eliminating y from the equations $y = 2 - 4x, \quad 3x^2 + xy + 11 = 0$, show that $x^2 - 2x - 11 = 0$.

b Hence, or otherwise, solve the simultaneous equations $y = 2 - 4x, \quad 3x^2 + xy + 11 = 0$, giving your answers in the form $a \pm b\sqrt{3}$, where a,b are integers.

Chapter 3 · Original slide 9 · Exercise 3A/B – Y1 Pages 40, 41

b Find the second pair of solutions for the simultaneous equations.

Chapter 3 · Original slide 10 · Extension

1

[MAT 2012 1G] There are *positive* real numbers x and y which solve the equations

$2x + ky = 4,$
$x + y = k$

for:

A) All values of k;

B) No values of k;

C) $k = 2$ only;

D) Only $k > -2$

2

[STEP 2010 Q1] Given that

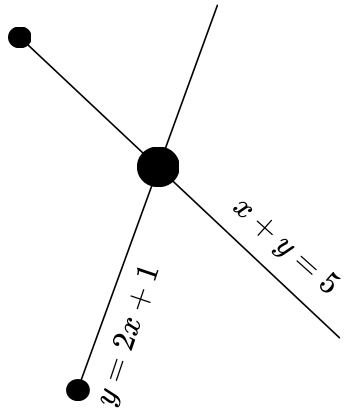
$5x^2 + 2y^2 - 6xy + 4x - 4y \equiv a(x - y + 2)^2 + b(cx + y)^2 + d$

a) Find the values of a, b, c, d.

b) Solve the simultaneous equations:

$5x^2 + 2y^2 - 6xy + 4x - 4y = 9,$
$6x^2 + 3y^2 - 8xy + 8x - 8y = 14$
(Hint: Can we use the same method in (a) to rewrite the second equation?)

Chapter 3 · Original slide 11 · Simultaneous Equations and Graphs



Now suppose we introduced a second simultaneous equation:

$y = 2x + 1$

$x + y = 5$

The second line again consists of all points (x,y) which satisfy the equation. So what point must satisfy both equations simultaneously?

Recall that a line with a given equation is the set of all points that satisfy the equation.

i.e. It is a graphical representation of the solution set where each (x, y) point represents each of the solutions x and y to the equation. e.g. $x = 3, y = 7$

Chapter 3 · Original slide 12 · Example

a) On the same axes, draw the graphs of:

$2x + y = 3$
$y = x^2 - 3x + 1$

b) Use your graph to write down the solutions to the simultaneous equations.

(We could always substitute into the original equations to check they work)

c) What algebraic method (perhaps thinking about the previous chapter), could we have used to show the graphs would have intersected twice?

Chapter 3 · Original slide 13 · Another Example

a) On the same axes, draw the graphs of:

$y = 2x - 2$
$y = x^2 + 4x + 1$

b) Prove algebraically that the lines never meet.

Chapter 3 · Original slide 14 · Final Example

a b

The line with equation $y = 2x + 1$ meets the curve with equation

$kx^2 + 2y + (k - 2) = 0$ at exactly one point. Given that k is a positive constant:

a) Find the value of k.

b) For this value of k, find the coordinates of this point of intersection.

Chapter 3 · Original slide 15 · Exercise 3C – Y1 Page 45

8 Given the simultaneous equations $2x - y = 1, \quad x^2 + 4ky + 5k = 0$, where k is a non-zero constant,

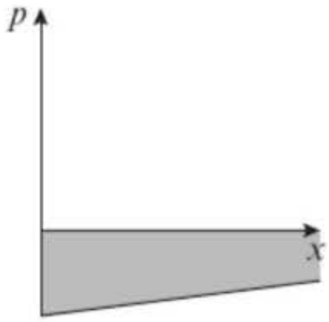
a show that $x^2 + 8kx + k = 0$.

Given that $x^2 + 8kx + k = 0$ has equal roots,

b find the value of k

c for this value of k, find the solution of the simultaneous equations.

Chapter 3 · Original slide 16 · Exercise 3C – Y1 Page 45



(5 marks)

9 A swimmer dives into a pool. Her position, p m, underwater can be modelled in relation to her horizontal distance, x m, from the point she entered the water as a quadratic equation $p = \frac{1}{2}x^2 - 3x$. The position of the bottom of the pool can be modelled by the linear equation $p = 0.3x - 6$. Determine whether this model predicts that the swimmer will touch the bottom of the pool.

Chapter 3 · Original slide 17 · Set Builder Notation

Recall that a set is a collection of values such that:

a) The order of values does not matter.

b) There are no duplicates.

$\{1, 2, 3\} \cap \{3, 4, 5\} =$

$\{1, 2, 3\} \cup \{3, 4, 5\} =$

$\{1, 2\} \cap \{3, 4\} =$

Recap from GCSE:

• We use curly braces to list the values in a set, e.g. $A = \{1, 4, 6, 7\}$

• If A and B are sets then $A \cap B$ is the intersection of A and B, giving a set which has the elements in A and B.

• $A \cup B$ is the union of A and B, giving a set which has the elements in A or in B.

• $\varnothing$ is the empty set, i.e. the set with nothing in it.

• Sets can also be infinitely large. $\mathbb{N}$ is the set of natural numbers (all positive integers), $\mathbb{Z}$ is the set of all integers (including negative numbers and 0) and $\mathbb{R}$ is the set of all real numbers (including all possible decimals).

• We write $x \in A$ to mean "x is a member of the set A". So $x \in \mathbb{R}$ would mean "x is a real number".

Chapter 3 · Original slide 18 · Set Builder Notation

It is possible to construct sets without having to explicitly list its values. We use:

The | or : means “such that”.

$\{expr|condition\}$
or $\{expr : condition\}$

Can you guess what sets the following give?

$\{2x : x \in \mathbb{Z}\} =$

$\{2^x : x \in \mathbb{N}\} =$

$\{xy : x, y \text{ are prime}\} =$

We previously talked about ‘solutions sets’, so set builder notation is very useful for specifying the set of solutions!

Chapter 3 · Original slide 19 · Set Builder Notation

Can you use set builder notation to specify the following sets?

All odd numbers.

All (real) numbers greater than 5.

All (real) numbers less than 5 or greater than 7.

All (real) numbers between 5 and 7 inclusive.

Chapter 3 · Original slide 20 · Recap of linear inequalities

Inequality	Solution Set
$2x + 1 > 5$	
$-x \geq 2$	

$2 \quad 3 \quad 4$

$3\,(x - 5) \geq 5 - 2(x - 8)$

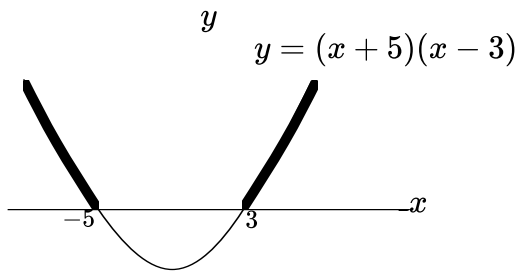
Note: Multiplying or both sides of an inequality by a negative number reverses the direction.

Combining Inequalities:

If $x < 3$ and $2 \leq x < 4$, what is the combined solution set?

If both inequalities have to be satisfied, we have to be on both lines. Place your finger vertically and scan across.

Chapter 3 · Original slide 21 · RECAP :: Solving Quadratic Inequalities



Solve $x^2 + 2x - 15 > 0$

Step 1: Get 0 on one side

$(x + 5)\,(x - 3) > 0$

Step 2: Factorise

Step 3: Sketch and reason

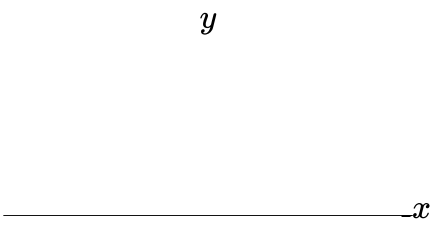
Since we sketched $y = (x + 5)(x - 3)$
we're interested where $y > 0$, i.e. the parts of the line where the y value is positive.

Click to bolden >

What can you say about the x values of points in this region?

What can you say about the x values of points in this region?

Chapter 3 · Original slide 22 · Solving Quadratic Inequalities



Solve $x^2 + 2x - 15 \leq 0$

Step 1: Get 0 on one side

Step 2: Factorise

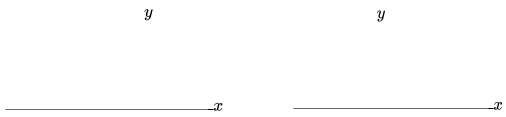
$(x + 5)(x - 3) \leq 0$

Step 3: Sketch and reason

Again, what can we say about the x value of any point in this region?

Note: we need $\leq$ rather than $<$ to be consistent with the original inequality.

Chapter 3 · Original slide 23 · Further Examples



Solve $x^2 + 5x \geq -4$

Solve $x^2 < 9$

Note: do not skip the ‘sketch step’.

To use the quadratic inequality solver on ClassWiz, go to Menu → Inequalities, then choose ‘order 2’ (i.e. quadratic)

Try the same on a CG50.

Chapter 3 · Original slide 24 · Test Your Understanding

Find the set of values of x for which

(a) $3(x-2) < 8-2x$, (2)

(b) $(2x-7)(1+x) < 0$, (3)

(c) both $3(x-2) < 8-2x$ and $(2x-7)(1+x) < 0$. (1)

Given that the equation $2qx^2 + qx - 1 = 0$, where q is a constant, has no real roots,

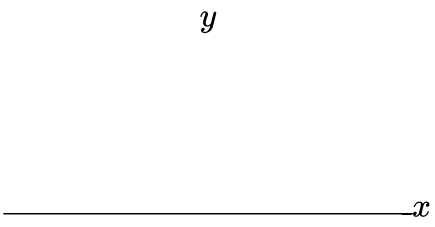
(a) show that $q^2 + 8q < 0$. (2)

(b) Hence find the set of possible values of q. (3)

Note: the original equation has no solutions, but the inequality $q^2 + 8q < 0$ *did* have solutions. But we have found the solutions for q that result in the original equation not having any solutions for x. These are different variables, so have different solutions sets, even if the solution set of q influences the solution set of x.

Chapter 3 · Original slide 25 · Deal with inequalities with a division by x

Find the set of values for which $\frac{6}{x} > 2, x \neq 0$



Why can't we just multiply both sides by x?

Chapter 3 · Original slide 26 · Exercise 3D/3E – Y1 Page 47-48, 50-51

3 Use set notation to describe the set of values of x for which:

g $4(3x + 7) \leq 20$ or $2(3x - 5) \geq \frac{7-6x}{2}$

6 Find the set of values of x for which $x^2 - 5x - 14 > 0$.

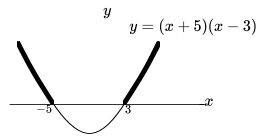
Chapter 3 · Original slide 27 · Exercise 3D/3E – Y1 Page 47-48, 50-51

8 Given that $x \neq 3$, find the set of values for which $\frac{5}{x-3} < 2$.

Chapter 3 · Original slide 28 · Exercise 3D/3E – Y1 Page 47-48, 50-51

9 The equation $kx^2 - 2kx + 3 = 0$, where k is a constant, has no real roots. Prove that k satisfies the inequality $0 \leq k < 3$.

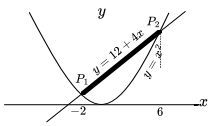
Chapter 3 · Original slide 29 · Inequalities on Graphs



When we solved quadratic inequalities, e.g. $(x + 5)(x - 3) > 0$

We plotted $y = (x + 5)(x - 3)$ and observed the values of x for which $y > 0$.

Can we use a similar method when we don't have 0 on one side?



Example: L_1 has equation $y = 12 + 4x$. L_2 has equation $y = x^2$.

The diagram shows a sketch of L_1 and L_2 on the same axes.

a) Find the coordinates of P_1 and P_2, the points of intersection.

b) Hence write down the solution to the inequality $12 + 4x > x^2$.

a b

Solve simultaneously to find points of intersection:

Chapter 3 · Original slide 30 · Inequality Regions

y

x

On graph paper, shade the region that satisfies the inequalities:

$2y + x < 14$
$y \geq x^2 - 3x - 4$

You did this at GCSE, the only difference here being that the graphs involved might not be straight lines.

Step 2:

An inequality involving x and y represents a 2D region in space.

Identify the correct side of each line each inequality represents.

Click to sketch >

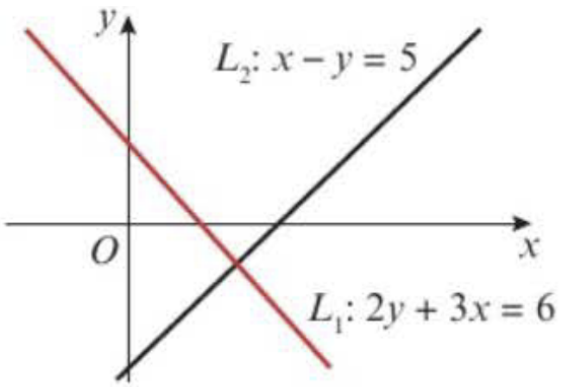
Tip: To quickly sketch $2y + x = 14$, consider what happens when x is 0 and when y is 0.

Tip: Make sure y is on the side where it is positive.

If y is on the smaller side, you're below the line.

If y is on the greater side, you're above the line.

Chapter 3 · Original slide 31 · Exercise 3F/3G – Y1 Page 53, 55



1 L_1 has equation $2y + 3x = 6$. L_2 has the equation $x - y = 5$. The diagram shows a sketch of L_1, L_2.

a Find the coordinates of P, the point of intersection.

b Hence write down the solution to the inequality $2y + 3x > x - y$.

Chapter 3 · Original slide 32 · Exercise 3F/3G – Y1 Page 53, 55

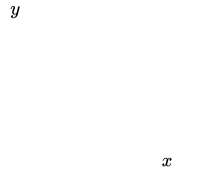
3 Find the set of values of x for which the curve with equation $y = \mathrm{f}(x)$ is below the line with equation $y = \mathrm{g}(x)$.

a $\mathrm{f}(x) = 3x^2 - 2x - 1$
$\mathrm{g}(x) = x + 5$

b $\mathrm{f}(x) = 2x^2 - 4x + 1$
$\mathrm{g}(x) = 3x - 2$

c $\mathrm{f}(x) = 5x - 2x^2 - 4$
$\mathrm{g}(x) = -2x - 1$

Chapter 4 · Original slide 21 · GCSE RECAP: Reciprocal Graphs





Sketch $y = \frac{1}{x}$

Sketch $y = -\frac{3}{x}$

Note: The scaling caused by the 3 isn't observable for this graph in isolation because the axes have no scale. This will only be observation for multiple graphs on the same axes.

Notice the distance between this line and the x-axis (i.e. the line $y = 0$) gradually decreases as the lines go off towards infinity. The line $y = 0$ is known as an asymptote of the graph.

! An asymptote is a line which the graph approaches but never reaches.

Asymptotes of $y = \frac{a}{x}$:

Chapter 4 · Original slide 22 · Reciprocal Graphs



Sketch $y = \frac{3}{x^2}$

Hint: Note that anything squared will always be at least 0.

This is new to the A Level 2017 syllabus.

Sketch $y = -\frac{4}{x^2}$

Chapter 4 · Original slide 23 · Reciprocal Graphs

On the same axes, sketch $y = \frac{1}{x}$ and $y = \frac{3}{x}$

$y = \frac{3}{x}$

$y = \frac{1}{x}$

y

The y value for $y = \frac{3}{x}$ will be 3 times greater than $y = \frac{1}{x}$

x

Chapter 4 · Original slide 24 · Exercise 4C – Pure Y1 page 67

1 Use a separate diagram to sketch each pair of graphs.

a $y = \frac{2}{x}$ and $y = \frac{4}{x}$ **b** $y = \frac{2}{x}$ and $y = -\frac{2}{x}$ **c** $y = -\frac{4}{x}$ and $y = -\frac{2}{x}$
d $y = \frac{3}{x}$ and $y = \frac{8}{x}$ **e** $y = -\frac{3}{x}$ and $y = -\frac{8}{x}$

Chapter 4 · Original slide 25 · Exercise 4C – Pure Y1 page 67

2 Use a separate diagram to sketch each pair of graphs.
a $y = \frac{2}{x^2}$ and $y = \frac{5}{x^2}$ **b** $y = \frac{3}{x^2}$ and $y = -\frac{3}{x^2}$ **c** $y = -\frac{2}{x^2}$ and $y = -\frac{6}{x^2}$

Chapter 5 · Original slide 43 · What's the point of straight line equations?







We saw in Chapter 2 that lots of things in real life have a 'quadratic' relationship, e.g. vertical height with time. Lots of real life variables have a 'linear' relationship, i.e. there is a fixed increase/decrease in one variable each time the other variable goes up by 1 unit.

Examples

Car sales made and take home pay.

(And a pure maths one:)

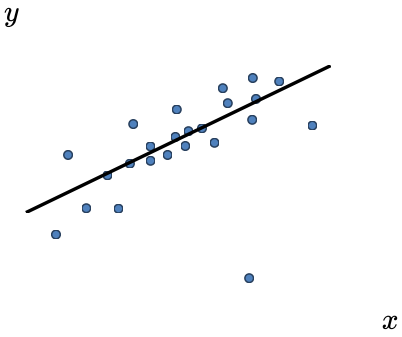
The nth term of an arithmetic series.

3, 5, 8, 11, 14, ...

The relationship between Celsius and Fahrenheit.

Temperature and altitude (in a particular location)

Chapter 5 · Original slide 44 · Modelling



100
80
60
40
20

A mathematical model is an attempt to model a real-life situation based on mathematical concepts.

For this example, our model might be a linear model with equation $\mathbf{y=mx+c}$ where x is a student's GCSE mark and y is the predicted A Level mark.

GCSE Maths mark

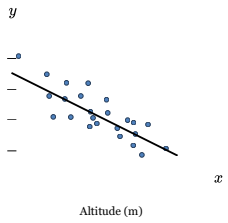
A Level Maths Mark

20 40 60 80 100

Such a linear model can be drawn as a line of best fit.

The data obviously doesn't fit this line exactly. This chosen model may only partially fit the data (and the further the points are away from the line, the less suitable this model is).

Chapter 5 · Original slide 45 · Example



100
80
60
40
20

Temperature $^\circ F$

50 100 150 200 250

a b c

The temperature y at different points on a mountain is recorded at different altitudes x.

Suppose we were to use a linear model $y = mx + c$.

Determine m and c (you can assume the line goes through $(0, 70)$ and $(250, 20)$.

Interpret the meaning of m and c in this context.

Predict at what altitude the temperature reaches $0^\circ F$.

Chapter 5 · Original slide 46 · Evaluating if a linear model is most suitable

Choosing a model for our data usually involves making simplifying assumptions. For a linear model we are assuming:

The y value goes up the same amount for each unit increase in x.

Note: Population tends to grow/decline by a percentage each year. Therefore a better model would be an exponential function $P = ab^t$ where a, b are constants. The a would be the initial population and the b the factor increase each year, e.g. $b = 1.1$ would be a 10% annual growth.

The current population of Bickerstonia is 26000. This year (2017) the population increased by 150. Matt decides to model the population P based on the years t after 2017 by the linear model:

$P = mt + c$

Why might this not be a suitable model?

Chapter 5 · Original slide 47 · Exercise 5H – Pure Y1 Page 106-108

3 The cost of electricity, E, in pounds and the number of kilowatt hours, h, are shown in the table.

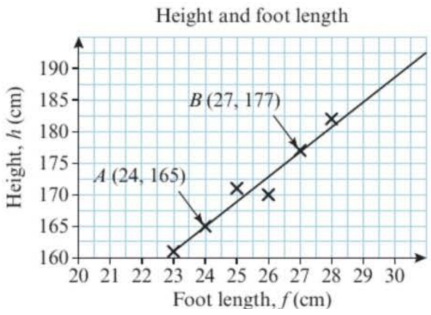
kilowatt hours, h	0	15	40	60	80	110
cost of electricity, E	45	46.8	49.8	52.2	54.6	58.2

a Draw a graph of the data.
b Explain how you know a linear model would be appropriate.
c Deduce an equation in the form $E = ah + b$.
d Interpret the meaning of the coefficients a and b.
e Use the model to find the cost of 65 kilowatt hours.

Chapter 5 · Original slide 48 · Exercise 5H – Pure Y1 Page 106-108

6 The average August temperature in Exeter is 20°C or 68°F. The average January temperature in the same place is 9°C or 48.2°F.
a Write an equation linking Fahrenheit F and Celsius C in the form $F = aC + b$. (3 marks)
b Interpret the values of a and b. (2 marks)
c The highest temperature recorded in the UK was 101.3°F. Calculate this temperature in Celsius.
d For what value is the temperature in Fahrenheit the same as the temperature in Celsius?

Chapter 5 · Original slide 49 · Exercise 5H – Pure Y1 Page 106-108



9 The price P of a good and the quantity Q of a good are linked.
The demand for a new pair of trainers can be modelled using the equation $P = -\frac{3}{4}Q + 35$.
The supply of the trainers can be modelled using the equation $P = \frac{2}{3}Q + 1$.
a Draw a sketch showing the demand and supply lines on the same pair of axes.
The equilibrium point is the point where the supply and demand lines meet.
b Find the values of P and Q at the equilibrium point.

8 The scatter graph shows the height h and foot length f of 8 students. A line of best fit is drawn on the scatter graph.
a Explain why the data can be approximated to a linear model. (1 mark)
b Use points A and B on the scatter graph to write a linear equation in the form $h = af + b$. (3 marks)
c Calculate the expected height of a person with a foot length of 26.5 cm. (1 mark)

Chapter 7 · Original slide 4 · 1: Simplifying Algebraic Fractions

Recall that you can simplify fractions by dividing the numerator and denominator by a common factor.

$\frac{x^2-1}{x^2+x} =$

Note: Do not leave 1 in the denominator!

$\frac{2x^2+11x+12}{x^2+9x+20} =$

Tip: $\frac{a-b}{b-a} = -1$

Hint: To identify common factors we need to factorise first.

$\frac{x^2+3x+2}{x+1} =$

Tip: Factorise the easier one first because it provides clues to the factorisation of the other.

$\frac{4-x^2}{x^2+2x-8} =$

Chapter 7 · Original slide 5 · Exercise 7A – Y1 Pages 138-139

2 Simplify these fractions as far as possible:

m $\frac{2x^2+3x+1}{x^2-x-2}$ **n** $\frac{x^2+6x+8}{3x^2+7x+2}$ **o** $\frac{2x^2-5x-3}{2x^2-9x+9}$

3 $\frac{6x^3+3x^2-84x}{6x^2-33x+42} = \frac{ax(x+b)}{x+c}$, where a, b and c are constants.
Work out the values of a, b and c.

Chapter 7 · Original slide 6 · Terminology

$11 \div 4 = 2rem3$

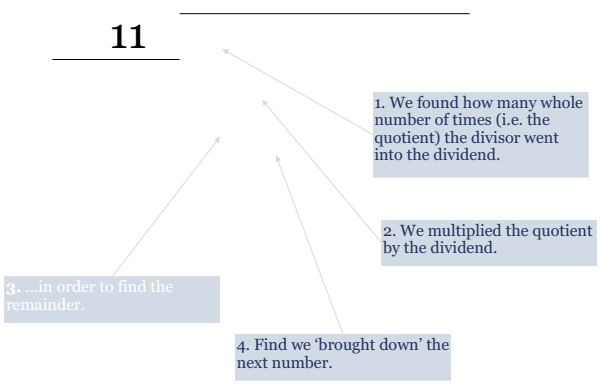
dividend
(the thing we're dividing)

quotient

divisor
(the thing we're dividing by)

remainder

Chapter 7 · Original slide 7 · Normal Long Division



4 2 3 . 0 0 0 0

Chapter 7 · Original slide 8 · Now repeat! How many times does x go into -2{x}^{2}?

$x + 5$

$$\begin{array}{r} 6x^3 + 30x^2 \\ \text{Fractions} \quad -\ 2x^2 - 7x \\ -\ 2x^2 - 10x \\ \hline 3x + 15 \\ 3x + 15 \\ \hline 0 \end{array}$$

Example

How many times does x go into $6x^3$?

$6x2$

- $2x$

+ 3

$6x3 + 28x2 - 7x + 15$

Subtract and carry down next term.

Multiply $6x^2$ by $(x + 5)$. The first term should match with above.

Tip:

You can check your solution by expanding:

$(x + 5)\left(6x^2 - 2x + 3\right)$

But if you know you should get no remainder, ending with 0 at the bottom is a good sign!

Chapter 7 · Original slide 9 · Further Example

x - 1

$3x3 + 0x2 - 2x + 4$

Find the remainder when $3x^3 - 2x + 4$ is divided by $x - 1$.

Note: Ensure that any missing terms in the polynomial are filled in, so that the powers decrease by 1 each time.

Chapter 7 · Original slide 10 · Test Your Understanding

Find the remainder when $2x^3 - 5x^2 - 16x + 10$ is divided by $x - 4$.

$x - 4$

$2x^3 - 5x^2 - 16x + 10$

Chapter 7 · Original slide 11 · Further Test Your Understanding

Divide $8x^3 - 1$ by $2x - 1$.

$8x^3 + 0x^2 + 0x - 1$

$2x - 1$

Fro Fun Fact: There is a well-known factorisation for the '*difference of two cubes*':

$x^3 - y^3 = (x - y)\left(x^2 + xy + y^2\right)$

This is NOT in the A Level syllabus.

Chapter 7 · Original slide 12 · Exercise 7B – Y1 Pages 142

15 $\mathrm{f}(x) = 10x^3 + 43x^2 - 2x - 10$.
Find the remainder when $\mathrm{f}(x)$ is divided by $(5x + 4)$.

16 $\mathrm{f}(x) = 3x^3 - 14x^2 - 47x - 14$.
a Find the remainder when $\mathrm{f}(x)$ is divided by $(x - 3)$.
b Given that $(x + 2)$ is a factor of $\mathrm{f}(x)$, factorise $\mathrm{f}(x)$ completely.

Chapter 7 · Original slide 13 · Exercise 7B – Y1 Pages 142

17 a Find the remainder when $x^3 + 6x^2 + 5x - 12$ is divided by
i $x - 2$,
ii $x + 3$.
b Hence, or otherwise, find all the solutions to the equation $x^3 + 6x^2 + 5x - 12 = 0$.

Chapter 7 · Original slide 14 · Exercise 7B – Y1 Pages 142

18 $\mathrm{f}(x) = 2x^3 + 3x^2 - 8x + 3$.
a Show that $\mathrm{f}(x) = (2x - 1)(ax^2 + bx + c)$ where a, b and c are constants to be found.
b Hence factorise $\mathrm{f}(x)$ completely.
c Write down all the real roots of the equation $\mathrm{f}(x) = 0$.

Chapter 7 · Original slide 15 · Exercise 7B – Y1 Pages 142

19 $\mathrm{f}(x) = 12x^3 + 5x^2 + 2x - 1$.
a Show that $(4x - 1)$ is a factor of $\mathrm{f}(x)$ and write $\mathrm{f}(x)$ in the form $(4x - 1)(ax^2 + bx + c)$.
b Hence, show that the equation $12x^3 + 5x^2 + 2x - 1 = 0$ has exactly 1 real solution.

Chapter 7 · Original slide 16 · 3: The Factor Theorem

$x^3 + x^2 - 4x - 4 = (x - 2)\left(x^2 + 3x + 2\right)$

We can see that $(x - 2)$ is a factor of $x^3 + x^2 - 4x - 4$.

What would happen if x is 2?

The converse is also true: if we could find a value a such that the LHS is 0 when we substitute in a for x, then $(x - a)$ would be a factor.

! The Factor Theorem states that if $f(x)$ is a polynomial then:

• If $f(p) = 0$, then $(x - p)$ is a factor of $f(x)$.

• Conversely, if $(x - p)$ is a factor of $f(x)$, then $f(p) = 0$.

Chapter 7 · Original slide 17 · Examples (in textbook)

Show that $(x-2)$ is a factor of x^3+x^2-4x-4.

Writing $f(x)=\ldots$ gives us appropriate notation, i.e. $f(2)$, to show we're substituting 2 into the polynomial on the next line.

Keep on trying values until you find one where $f(p)=0$. Recommended order: $p=1,-1,2,-2,\ldots$

You can use the 'Table' mode on your calculator to try lots of values in a range – see my interactive Classwiz powerpoint.

Tip: We can determine b by considering for example the x terms in the expansion. This is much faster than algebraic division.

Let $f(x)=x^3+x^2-4x-4$

Fully factorise $2x^3+x^2-18x-9$.

Let $f(x)=2x^3+x^2-18x-9$

Chapter 7 · Original slide 18 · Using Factor Theorem to find unknown coefficients

Given that $2x+1$ is a factor of $6x^3+ax^2+1$, determine the value of a.

Notes:

Choose the value of x to substitute that makes the divisor/factor 0.

So if $2x+1=0$, then $x=-\frac{1}{2}$, thus find $f\left(-\frac{1}{2}\right)$.

This even works when you're dividing by non-linear factors, e.g. $\frac{1}{x}+1$, but at A Level they will always be of the form $ax+b$.

Chapter 7 · Original slide 19 · Test Your Understanding

$\mathrm{f}(x) = 6x^3 + 13x^2 - 4$

(a) Use the remainder theorem to find the remainder when $\mathrm{f}(x)$ is divided by $(2x + 3)$. (2)
(b) Use the factor theorem to show that $(x + 2)$ is a factor of $\mathrm{f}(x)$. (2)
(c) Factorise $\mathrm{f}(x)$ completely. (4)

Given that $3x - 1$ is a factor of $3x^3 + 11x^2 + ax + 1$, determine the value of a.

No long in spec.

Chapter 7 · Original slide 20 · Exercise 7C – Y1 Pages 146

13 $\mathrm{f}(x) = 3x^3 - 12x^2 + 6x - 24$.
a Use the factor theorem to show that $(x - 4)$ is a factor of $\mathrm{f}(x)$.
b Hence, show that 4 is the only real root of the equation $\mathrm{f}(x) = 0$.

Chapter 7 · Original slide 21 · Exercise 7C – Y1 Pages 145-146

14 $\mathrm{f}(x) = 4x^3 + 4x^2 - 11x - 6$.
a Use the factor theorem to show that $(x + 2)$ is a factor of $\mathrm{f}(x)$.
b Factorise $\mathrm{f}(x)$ completely.

Chapter 7 · Original slide 22 · Exercise 7C – Y1 Pages 145-146

15 a Show that $(x - 2)$ is a factor of $9x^4 - 18x^3 - x^2 + 2x$.
b Hence, find four real solutions to the equation $9x^4 - 18x^3 - x^2 + 2x = 0$.

Chapter 7 · Original slide 23 · Exercise Extension

1 2

[MAT 2006 1E] The cubic

$x^3 + ax + b$

Has both $(x - 1)$ and $(x - 2)$ as factors. Determine the values of a and b.

[MAT 2009 1I] The polynomial

$n^2x^{2n+3} - 25nx^{n+1} + 150x^7$

Has $x^2 - 1$ as a factor

A) for no values of n;

B) for $n = 10$ only;

C) for $n = 15$ only;

D) for $n = 10$ and $n = 15$ only.

3

[MAT 2013 1G] Let $n \geq 2$ be an integer and $p_n(x)$ be the polynomial

$p_n(x) = (x - 1) + (x - 2) + \ldots + (x - n)$

What is the remainder, in terms of n, when $p_n(x)$ is divided by $p_{n-1}(x)$?

The remainder theorem states that if $f(x)$ is divided by $(x - a)$, the remainder is $f(a)$. This similarly works whenever a makes the divisor 0.

Chapter 8 · Original slide 4 · Patterns in binomial expansions

a)

Expand $(a + b)^0$.

b)

Expand $(a + b)^1$.

c)

Expand $(a + b)^2$.

d)

Expand $(a + b)^3$.

e)

Expand $(a + b)^4$.

What do you notice?

Describe the coefficients and the powers of a and b.

Chapter 8 · Original slide 5 · Pascal’s triangle

Each entry, except the outside 1s, is the sum of the two entries immediately above it.

Memorise these first rows.

Chapter 8 · Original slide 6 · Expanding (2 + 3x)⁴

Find the expansion of $(2+3x)^4$.

1. Use row 4 of Pascal’s triangle.

Chapter 8 · Original slide 7 · Expanding a bracket with subtraction

Expand $(1-2x)^3$.

Write the bracket as $1+(-2x)$. The second term is $-2x$.

Chapter 8 · Original slide 8 · Finding a single coefficient

The coefficient of x^2 in $(2-cx)^5$ is 720. Find the possible values of c.

Chapter 8 · Original slide 9 · Test your understanding

(a) Find the first three terms, in ascending powers of x, of $(2 + kx)^7$, where k is a constant. Give each term in its simplest form. (4 marks)
Given that the coefficient of x^2 is six times the coefficient of x, (b) find k. (2 marks)

Chapter 8 · Original slide 10 · Exercise 8A · Questions 4 and 5

Question 4

4 Fully expand the expression $(1 + 3x)(1 + 2x)^3$.

Question 5

5 Expand $(2 + y)^3$. Hence or otherwise, write down the expansion of $(2 + x - x^2)^3$ in ascending powers of x.

Chapter 8 · Original slide 11 · Exercise 8A · Questions 6 and 7

Question 6

6 The coefficient of x^2 in the expansion of $(2+ax)^3$ is 54. Find the possible values of the constant a.

Question 7

7 The coefficient of x^3 in the expansion of $(2-x)(3+bx)^3$ is 45. Find possible values of the constant b.

Chapter 8 · Original slide 12 · Exercise 8A – Y1 page 160-161

8 Work out the coefficient of x^2 in the expansion of $(p-2x)^3$. Give your answer in terms of p.

9 After 5 years, the value of an investment of \pounds500 at an interest rate of $X\%$ per annum is given by

$$500\left(1+\frac{X}{100}\right)^5$$

Find an approximation for this expression in the form $A+BX+CX^2$, where A, B and C are constants to be found. You can ignore higher powers of X.

Chapter 8 · Original slide 13 · Extension · A binomial equation

How many pairs of positive integers x, y satisfy $x^3 + 6x^2y + 12xy^2 + 8y^3 = 2^{30}$?

A) 0
B) 2^6
C) $2^9 - 1$
D) $2^{10} + 2$
E) — (blank in the supplied slide)

Chapter 8 · Original slide 14 · Factorial and the choose function

$n! = n(n-1)\cdots 2 \cdot 1$
Say “n factorial”. It counts the ways of arranging n distinct objects in a line.

$\binom{n}{r} = \frac{n!}{r!(n-r)!}$
Say “n choose r”. It counts selections of r objects from n, when order does not matter.

Three letters A, B and C have $3 \times 2 \times 1 = 3! = 6$ arrangements. Use the factorial button on your calculator.

Choosing four people from ten gives $\binom{10}{4} = \frac{10!}{4!6!} = 210$ selections. Use the nCr button. The $\binom{n}{r}$ notation is preferable in written work.

Chapter 8 · Original slide 15 · Factorial and choose examples

Calculate these values using the factorial button, without the nCr button.

a)
$5!$

b)
$\binom{5}{3}$

c)
$0!$

d)
$\binom{20}{1}$

e)
$\binom{20}{0}$

f)
$\binom{20}{2}$

g)
$\binom{20}{18}$

Chapter 8 · Original slide 16 · Binomial coefficients and Pascal's triangle

Number the top row 0 and the first entry 0. Entry k in row n is $\binom{n}{k}$. The top number gives the row. The bottom number gives the entry.

For large powers, such as $(x+3)^{20}$, calculate coefficients with the choose function rather than building all the rows.

Chapter 8 · Original slide 17 · Pascal's rule

This is useful enrichment for STEP. You are not required to know the proof for this course.

Chapter 8 · Original slide 18 · Exercise 8B – Y1 page 162

10 Given that $\binom{50}{13} = \frac{50!}{13!a!}$, write down the value of a.

11 Given that $\binom{35}{p} = \frac{35!}{p!18!}$, write down the value of p.

Chapter 8 · Original slide 19 · Exercise 8B – Y1 page 162

Challenge
a Work out ${}^{10}C_3$ and ${}^{10}C_7$.
b Work out ${}^{14}C_5$ and ${}^{14}C_9$.
c What do you notice about your answers to parts **a** and **b**?
d Prove that ${}^{n}C_r = {}^{n}C_{n-r}$.

Chapter 8 · Original slide 20 · Why binomial coefficients count the terms

The entries in row 5 are $1, 5, 10, 10, 5, 1$, or $\binom{5}{0}, \binom{5}{1}, \ldots, \binom{5}{5}$.

$(a+b)^5 = (a+b)(a+b)(a+b)(a+b)(a+b)$

Every term comes from choosing one term from each bracket. How many choices produce a^3b^2?

Chapter 8 · Original slide 21 · The binomial expansion formula

$$(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$$

For n a non-negative integer, each term is $\binom{n}{r} a^{n-r} b^r$. The two powers add up to n. Fractional and negative powers are studied in Year 2.

Example

Find the first four terms of $(3x+1)^{10}$, in ascending powers of x.

Chapter 8 · Original slide 22 · Test your understanding

First three terms

Find the first three terms of $(2 - \frac{1}{3}x)^7$, in ascending powers of x.

Chapter 8 · Original slide 23 · Exercise 8C – Y1 page 164

5 Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(2-x)^6$ and simplify each term.

6 Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(3-2x)^5$ giving each term in its simplest form.

7 Find the binomial expansion of $\left(x+\frac{1}{x}\right)^5$ giving each term in its simplest form.

Chapter 8 · Original slide 24 · Extension

1

[AEA 2013 Q1a] In the binomial expansion of $\left(1+\frac{12n}{5}x\right)^n$ the coefficients of x^2 and x^3 are equal and non-zero.

Find the possible values of n.

Hint: Remember that $\binom{n}{2} = \frac{n(n-1)}{2!}$

Can you similarly simplify $\binom{n}{3}$ using $\binom{n}{r} = \frac{n!}{r!(n-r)!}$?

2

[STEP I 2010 Q5a] By considering the expansion of $(1+x)^n$, where n is a positive integer, or otherwise, show that:

$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \ldots + \binom{n}{n} = 2^n$

reflection: This means that the sum of each row in Pascal's Triangle gives successive powers of 2. Safe!

Chapter 8 · Original slide 25 · Selecting a general term

$T_{r+1} = \binom{n}{r} a^{n-r} b^r$

Expression	Power of x wanted	Required term
$(a+x)^{10}$	3	
$(2x-1)^{75}$	50	
$(3-x)^{12}$	7	
$(3x+4)^{16}$	3	

The powers of the two terms always add up to the power of the original bracket.

Chapter 8 · Original slide 26 · Finding a coefficient

The coefficient of x^4 in $(1+qx)^{10}$ is 3360. Find the possible values of q.

Chapter 8 · Original slide 27 · Comparing coefficients

Test your understanding

In $(1 + ax)^{10}$, a is non-zero and the coefficient of x^3 is twice the coefficient of x^2. Find a.

Chapter 8 · Original slide 28 · Exercise 8D · Question 5

When $(1 - 2x)^p$ is expanded, the coefficient of x^2 is 40. Given that $p > 0$, find (a) p, (b) the coefficient of x, and (c) the coefficient of x^3.

Chapter 8 · Original slide 29 · Exercise 8D – Y1 page 166-167

6 a Find the first three terms, in ascending powers of x, of the binomial expansion of $(5 + px)^{30}$, where p is a non-zero constant.
b Given that in this expansion the coefficient of x^2 is 29 times the coefficient of x work out the value of p.

Chapter 8 · Original slide 30 · Exercise 8D – Y1 page 166-167

7 a Find the first four terms, in ascending powers of x, of the binomial expansion of $(1 + qx)^{10}$, where q is a non-zero constant.
b Given that in the expansion of $(1 + qx)^{10}$ the coefficient of x^3 is 108 times the coefficient of x, work out the value of q.

Chapter 8 · Original slide 31 · Exercise 8D – Y1 page 166-167

8 a Find the first three terms, in ascending powers of x, of the binomial expansion of $(1 + px)^{11}$, where p is a constant.
b The first 3 terms in the same expansion are $1, 77x$ and qx^2, where q is a constant. Find the value of p and the value of q.

Chapter 8 · Original slide 32 · Exercise 8D – Y1 page 166-167

9 a Write down the first three terms, in ascending powers of x, of the binomial expansion of $(1 + px)^{15}$, where p is a non-zero constant.
b Given that, in the expansion of $(1 + px)^{15}$, the coefficient of x is $(-q)$ and the coefficient of x^2 is $5q$, find the value of p and the value of q.

Chapter 8 · Original slide 33 · Exercise 8D · Question 10

In $(1+x)^{30}$, the coefficients of x^9 and x^{10} are p and q. Find q/p.

Chapter 8 · Original slide 34 · Extension

1

[MAT 2014 1G] Let n be a positive integer. The coefficient of x^3y^5 in the expansion of $\left(1+xy+y^2\right)^n$ equals:

A) n

B) 2^n

C) $\binom{n}{3}\binom{n}{5}$

D) $4\binom{n}{4}$

E) $\binom{n}{8}$

2

[STEP I 2013 Q6] By considering the coefficient of x^r in the series for $(1+x)(1+x)^n$, or otherwise, obtain the following relation between binomial coefficients:

$\binom{n}{r}+\binom{n}{r-1}=\binom{n+1}{r}$

Chapter 8 · Original slide 35 · Estimating powers

(a) Find the first four terms of $(1+\frac{x}{4})^8$, in ascending powers of x, giving each term in its simplest form. (4 marks)
(b) Use the expansion to estimate 1.025^8 to four decimal places. (3 marks)

Chapter 8 · Original slide 36 · Test your understanding · Estimation

(a) Find the first four terms of $(1+\frac{x}{2})^{10}$, in ascending powers of x, in simplest form. (4 marks)
(b) Use the expansion to estimate 1.005^{10} to five decimal places. (3 marks)

Chapter 8 · Original slide 37 · Exercise 8E · Question 7

(a) Find the first three terms of $(3 - \frac{x}{5})^{10}$, in ascending powers of x, in simplest form. (4 marks)
(b) Explain how to use the expansion to estimate 2.98^{10}. (1 mark)

Chapter 8 · Original slide 38 · Exercise 8E · Question 8

(a) Find the first four terms of $(1 - 3x)^5$, in ascending powers of x, in simplest form. (4 marks)
(b) For small x, ignore x^2 and higher powers and show that $(1 + x)(1 - 3x)^5 \approx 1 - 14x$. (2 marks)

Chapter 8 · Original slide 39 · Exercise 8E – Y1 Page 168-169

9 A microchip company models the probability of having no faulty chips on a single production run as:

$$\mathrm{P}(\text{no fault}) = (1-p)^n,\ p < 0.001$$

where p is the probability of a single chip being faulty, and n being the total number of chips produced.

a State why the model is restricted to small values of p. (1 marks)

b Given that $n = 200$, find an approximate expression for $\mathrm{P}(\text{no fault})$ in the form $a + bp + cp^2$. (2 marks)

c The company wants to achieve a 92% likelihood of having no faulty chips on a production run of 200 chips. Use your answer to part **b** to suggest a maximum value of p for this to be the case. (4 marks)

Chapter 10 · Original slide 29 · Quadratics in sin/cos/tan

We saw that an equation can be 'quadratic in' something, e.g. $x - 2\sqrt{x} + 1 = 0$ is 'quadratic in $\sqrt{x}$', meaning that $\sqrt{x}$ could be replaced with another variable, say y, to produce a quadratic equation $y^2 - 2y + 1 = 0$.

Solve $5\sin^2 x + 3\sin x - 2 = 0$ in the interval $0 \le x \le 360^\circ$.

Method 1: Use a substitution.

Method 2: Factorise without substitution.

This is the same, but we 'imagine' $\sin x$ as a single variable and hence factorise immediately.

Chapter 10 · Original slide 30 · More Examples

Solve $\tan^2\theta = 4$ in the interval $0 \leq x \leq 360°$.

Missing the negative case would result in the loss of multiple marks. Beware!

Solve $2\cos^2 x + \sin x = 3\sin^2 x$ in the interval $-180° \leq x \leq 180°$.

Tip: We have an identity involving sin^2 and cos^2, so it makes sense to change the squared one that would match all the others.

Chapter 10 · Original slide 31 · Test Your Understanding

(a) Show that the equation $5\sin x = 1 + 2\cos^2 x$ can be written in the form
$2\sin^2 x + 5\sin x - 3 = 0$. (2)
(b) Solve, for $0 \leq x < 360°$,
$2\sin^2 x + 5\sin x - 3 = 0$. (4)

Chapter 10 · Original slide 32 · Exercise 10F – Y1 P221-222

5 Find all the solutions, in the interval $0 \leq x \leq 360^\circ$, to the equation $8\sin^2 x + 6\cos x - 9 = 0$ giving each solution to one decimal place. (6 marks)

6 Find, for $0 \leq x \leq 360^\circ$, all the solutions of $\sin^2 x + 1 = \frac{7}{2}\cos^2 x$ giving each solution to one decimal place.

Chapter 10 · Original slide 33 · Exercise 10F – Y1 P221-222

7 Show that the equation $2\cos^2 x + \cos x - 6 = 0$ has no solutions.

b Hence show that the equation $\cos^2 x = 2 - \sin x$ has no solutions. (3 marks)

8 a Show that the equation $\cos^2 x = 2 - \sin x$ can be written as $\sin^2 x - \sin x + 1 = 0$. (2 marks)

Chapter 10 · Original slide 34 · Exercise 10F – Y1 P221-222

9 $\tan^2 x - 2\tan x - 4 = 0$.
a Show that $\tan x = p \pm \sqrt{q}$ where p and q are numbers to be found.
b Hence solve the equation $\tan^2 x - 2\tan x - 4 = 0$ in the interval $0 \leq x \leq 540^\circ$.

Chapter 14 · Original slide 42 · Quadratics in {e}^{x}

In previous chapters we've already dealt with quadratics in disguise, e.g. "quadratic in sin". We therefore just apply our usual approach: either make a suitable substitution so the equation is then quadratic, or (strongly recommended!) go straight for the factorisation.

Solve $e^{2x} + 2e^x - 15 = 0$

Solve $e^x - 2e^{-x} = 1$

Note that $e^{2x} = (e^x)^2$ therefore

First write negative powers as fractions:

Chapter 14 · Original slide 43 · Test Your Understanding

Solve $\ln(3x+1) = 2$

Solve $2^x e^{x+1} = 3$ giving your answer as an exact value.

Solve $e^{2x} + 5e^x = 6$

Chapter 14 · Original slide 44 · Exercise 14G – Y1 Page 327

4 Find the exact solutions to the equation $e^x + 12e^{-x} = 7$.

6 Solve $3^x e^{4x-1} = 5$, giving your answer in the form $\dfrac{a + \ln b}{c + \ln d}$